

PulseEDU Migration Guide

Feature Enablement + Staff Pilots — shipped July 2, 2026

For: development team | Scope: database schema, server enforcement, admin UI

1. What shipped (summary)

Nine modules that were previously always-on are now individually licensable, with two layers of control and an optional per-staff pilot mode:

- Modules: Data Chats, Parent Pick-Up, Event Ticketing, School Tours, E-Sign, Brain Lab, Gradebook, School Grade Calculator, Safety Plans.
- District switch (`super_feature_*`): set by SuperUser / plan licensing. Always wins — nothing below can override it.
- School switch (`feature_*`): set by school admin in Settings > School Features.
- Staff pilots (new): with the school switch OFF and the district switch ON, an admin can enable a feature for a hand-picked list of staff (`staff_feature_pilots` table).
- ALL switches default to TRUE. Nothing changes for existing schools until someone flips a toggle.

Effective enablement is computed in exactly one place — `loadEffectiveFeatures()` in `artifacts/api-server/src/lib/featureLicensing.ts`:

```
enabled = superOn && (schoolOn || actorHasPilotRow)
```

2. Database changes

2.1 How migrations run in this project

There is NO drizzle-kit migration file. All DDL is idempotent boot-time SQL in `artifacts/api-server/src/seed.ts` (`ALTER TABLE ... IF NOT EXISTS` pattern). It runs automatically every time the API server starts, so a fresh environment self-migrates on first boot. If you need to apply it manually to another database, the equivalent SQL is below.

2.2 `school_settings` — 18 new boolean columns (all NOT NULL DEFAULT TRUE)

```
-- one pair per module; <col> in: data_chats, pickup, ticketing, tours,
--   esign, brain_lab, gradebook, school_grade, safety_plans
ALTER TABLE school_settings
  ADD COLUMN IF NOT EXISTS feature_<col>          BOOLEAN NOT NULL DEFAULT TRUE;
ALTER TABLE school_settings
  ADD COLUMN IF NOT EXISTS super_feature_<col>     BOOLEAN NOT NULL DEFAULT TRUE;
```

2.3 New table: `staff_feature_pilots`

```
CREATE TABLE IF NOT EXISTS staff_feature_pilots (
  id SERIAL PRIMARY KEY,
  school_id INTEGER NOT NULL,
  feature_key TEXT NOT NULL, -- validated against FEATURE_KEYS in API
  staff_id INTEGER NOT NULL,
  granted_by_staff_id INTEGER,
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
CREATE UNIQUE INDEX IF NOT EXISTS staff_feature_pilots_unique
ON staff_feature_pilots (school_id, feature_key, staff_id);
```

Drizzle schema lives in lib/db/src/schema/staffFeaturePilots.ts; the 18 columns are in lib/db/src/schema/schoolSettings.ts.

2.4 plans JSONB backfill (only-if-absent)

Every existing plan's features JSONB gets the 9 new keys set to true, but ONLY if the key is absent. Without this, the next plan re-apply would flip the formerly-always-on modules OFF for every school on a plan. COALESCE keeps it idempotent and respects a district later deliberately setting a key to false.

```
UPDATE plans SET features = features || jsonb_build_object(
  'dataChats', COALESCE(features->'dataChats', 'true'::jsonb),
  'pickup', COALESCE(features->'pickup', 'true'::jsonb),
  'ticketing', COALESCE(features->'ticketing', 'true'::jsonb),
  'tours', COALESCE(features->'tours', 'true'::jsonb),
  'esign', COALESCE(features->'esign', 'true'::jsonb),
  'brainLab', COALESCE(features->'brainLab', 'true'::jsonb),
  'gradebook', COALESCE(features->'gradebook', 'true'::jsonb),
  'schoolGrade', COALESCE(features->'schoolGrade', 'true'::jsonb),
  'safetyPlans', COALESCE(features->'safetyPlans', 'true'::jsonb)
);
```

3. Server changes

3.1 Feature registry — lib/featureLicensing.ts

- FEATURE_KEYS gained 9 FeatureSpec entries (keys: dataChats, pickup, ticketing, tours, esign, brainLab, gradebook, schoolGrade, safetyPlans).
- gradebook declares requires: dataImports (dependency surfaces already enforce this).
- pilotable:false on all family-facing keys: ticketing, parentPortal, familyComm, schoolStoreNotify. Rationale: parents have no staff identity, so a pilot would silently gate families.
- loadEffectiveFeatures() unions the requesting staff member's pilot rows — a pilot row only matters when the school toggle is OFF and the district super toggle is ON.

3.2 Route enforcement — routes/index.ts

requireFeature(<key>) middleware is mounted in front of each module router. It returns 404 (looks-like-not-exist) when the feature is off, keeping the API surface dark. Public/unauthenticated surfaces are deliberately left OPEN:

- /ticketing/scan (no-login volunteer scanning)
- /esign/sign/:token (family signing links)
- /tours/public/* and /tours/walk/* (public brag page + live tour walk)
- Parent-portal routes use requireFeatureForParent (school-level check, no pilots): /parent/tickets, /parent/brain-lab.

3.3 Settings write path — routes/schoolSettings.ts

- PUT /api/school-settings accepts the 9 new camelCase keys (featureDataChats, featurePickup, featureTicketing, featureTours, featureEsign, featureBrainLab, featureGradebook, featureSchoolGrade, featureSafetyPlans) through the existing generic FEATURE_KEYS loop — same gating as older feature keys.

3.4 New pilot APIs — routes/featureLicensing.ts

```
GET /api/school-features/pilots
  -> { <featureKey>: [staffId, ...] }   (admin or SuperUser at acting school)

PUT /api/school-features/pilots/:featureKey  body: { staffIds: number[] }
  -> full-replace (transaction: delete + insert)
  400 unknown-or-non-pilotable key; staffIds must be ACTIVE staff at
  req.schoolId (cross-school ids rejected); 403 for non-admins
```

3.5 Gradebook goes dark end-to-end (important)

Because Gradebook has data-mutating side doors and embedded read surfaces, turning it off closes ALL of them (a code review caught rollback as a bypass; it is fixed):

- Importer preview + commit: requireFeature('gradebook') (routes/dataImports.ts).
- Rollback: POST /data-imports/jobs/:id/rollback returns 404 for gradebook-kind jobs when the feature is off.
- Job history: GET /data-imports/jobs returns 404 for ?kind=gradebook and silently omits gradebook rows from unfiltered listings.
- Embedded grade reads: loadCurrentGrades() in lib/studentMetrics.ts returns no rows unless BOTH school-level switches are on. This darkens Student Profile, Student Snapshot, parent portal Academics, HeartBEAT PDF, insights drill-downs, exports, and data-chat D/F scoping in one place.

Design decision: pilots do NOT re-enable embedded reads — the shared loader has no staff context and also feeds parent-facing surfaces. Pilots unlock the importer workflow routes only.

4. Client changes

- artifacts/client/src/App.tsx: hoisted useFeatureVisible hooks for the new keys; nav items, Quick Access tiles, Settings tile cards, and section renders are gated (dataChats, safetyPlans, pulseBrainLab, pickup, tours, ticketing, esign, schoolGrade; gradebook has no nav item of its own).
- School Features grid (Settings): +9 rows; Ticketing row is marked family-facing / not pilotable.
- NEW component artifacts/client/src/components/FeaturePilotsPanel.tsx: renders under the School Features save button; per-feature staff search + checkbox picker; PUT full-replace per key; refreshes effective features after save.

Client gating is UX only. Every gate is enforced server-side — never rely on the client checks for security.

5. Invariants your team must preserve

- Effective enablement is computed ONLY in loadEffectiveFeatures(). Never re-implement the superOn/schoolOn/pilot math at a call site.
- District license always wins: no school toggle or pilot row may enable a feature whose super_feature_* is false.
- Family-facing feature keys must be pilotable:false; the pilot PUT must keep rejecting them.
- A licensed module must go dark on EVERY surface together: route mounts, mutation side-doors (e.g. rollback), job/history listings, and embedded reads. Gradebook is the reference implementation.
- All new switches default TRUE — any future key added to FEATURE_KEYS needs the same seed.ts column-

add + plans JSONB only-if-absent backfill, or the next plan re-apply will turn the module off for everyone.

6. How it was verified (repeatable)

End-to-end curl tests against the dev API (all passing):

- Defaults: admin sees all 9 keys enabled in GET /api/me/features — behavior unchanged.
- School toggle off -> staff hits 404 on a module route (e.g. /api/data-chats/reminder).
- Pilot: PUT pilots/dataChats {staffIds:[teacher]} -> that teacher gets 200 while the school toggle stays off; a non-pilot admin still gets 404.
- District wins: super switch off + pilot row present -> still 404.
- Ticketing pilot PUT -> 400; unknown key -> 404; pilots GET as teacher -> 403; cross-school staffIds rejected.
- Gradebook dark: feature off -> rollback 404, jobs?kind=gradebook 404, unfiltered jobs list contains 0 gradebook rows, Student Profile shows 0 grade rows; all restored when re-enabled.
- Typechecks pass: npm run typecheck (api-server + client).

7. File map (everything touched today)

```
DB schema
  lib/db/src/schema/schoolSettings.ts      +18 boolean columns
  lib/db/src/schema/staffFeaturePilots.ts   NEW table schema
Server
  artifacts/api-server/src/seed.ts          idempotent DDL + plans backfill
  artifacts/api-server/src/lib/featureLicensing.ts +9 specs, pilot union
  artifacts/api-server/src/routes/featureLicensing.ts GET/PUT pilots
  artifacts/api-server/src/routes/index.ts   requireFeature mounts
  artifacts/api-server/src/routes/schoolSettings.ts +9 PUT keys
  artifacts/api-server/src/routes/dataImports.ts gradebook gates
  artifacts/api-server/src/lib/studentMetrics.ts read-surface gate
Client
  artifacts/client/src/App.tsx              nav/tile/section gating
  artifacts/client/src/components/FeaturePilotsPanel.tsx NEW
  artifacts/client/src/lib/features.tsx     feature hook plumbing
Docs
  docs/shipped.md                          shipped entry (detail)
  replit.md                                product bullet + invariants
```

8. Deploying this

- No manual DB step is required: the seed.ts DDL runs on API server boot and is idempotent, so publishing the app migrates production automatically on first start.
- If you manage a separate database by hand, run the SQL from section 2 (safe to re-run).
- Rollout is zero-risk by default: every switch ships TRUE, so behavior is identical until an admin or the district flips a toggle.